

Blessed International Learning Center Saigon

LOCK_HYBRID_AI

Face and RFID Access Control & Attendance System

Edge AI Node: **NXP FRDM-MCXN947** · Cloud Backend: **FastAPI + SQLite**

Technical Overview · Edge AI on the door, recognition in the cloud

The problem, and what the system is for

⚠ Why this system

- Manual attendance is slow and easy to cheat – one person signs in for another.
- A card on its own gets lost, lent out, or cloned – the UID is not a secret.
- Identification must be fast, accurate, and must survive a network outage.

Approach	Limitation	Consequence
Card only	UID can be cloned (SR-11)	Weak as a sole credential
Single biometric	Needs contact, slow in a queue	Poor experience at a door

◎ Design goal

Combine **face and card**: unlock in **under 2 s** by face or **under 0.5 s** by card, keep working offline, and keep the latch **fail-locked** under every fault.

LOCK_HYBRID_AI in one line

Concept

A system in **two halves**: the device at the door answers *“is someone there?”*, the server answers *“who is it?”* – because the processor at the door can find a face, but cannot run the much larger recognition model, and cannot hold the user database.

Edge Node

Detects faces on the integrated NPU, reads RFID cards, drives the door latch.

Transport

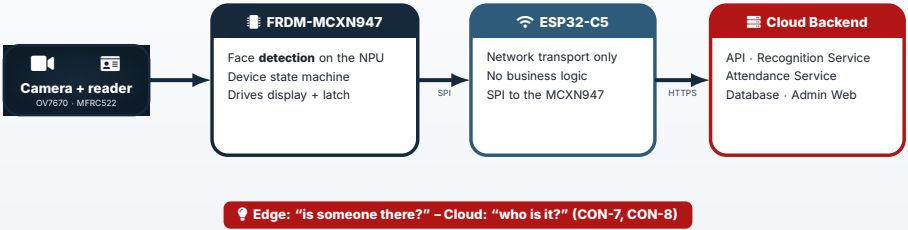
ESP32-C5 carries the face image or card UID to the server over Wi-Fi.

Cloud Backend

Identifies the face, stores attendance records, serves the admin web app.

Fail-locked by default | **Cards work offline** | **No face image kept on the device**

System architecture



Hardware and bill of materials

Component	Role / Specification
Controller	NXP FRDM-MCXN947 (integrated NPU)
Camera	OV7670 (DVP capture via SmartDMA)
RFID reader	MFRC522 (13.56 MHz)
Display	2.4 inch SPI TFT (240×320)
Latch actuator	Servo motor (defaults to locked – SR-6)
Network module	ESP32-C5 DevKit 2.0 (Wi-Fi)
Server	Cloud Backend – FastAPI + SQLite

Hardware design rule

Every peripheral connects to the controller and to nothing else – no peripheral talks to another peripheral directly.

Flow: attendance by face



Timing requirement

Target **under 2 seconds** from presenting a face to the latch beginning to open (PR-1). No decision within 5 s → treated as denied (FR-18).

Privacy

The captured face is erased from the device as soon as it has been sent – it is never written to flash or the SD card (SR-3).

Cards and offline operation

Decided on the device

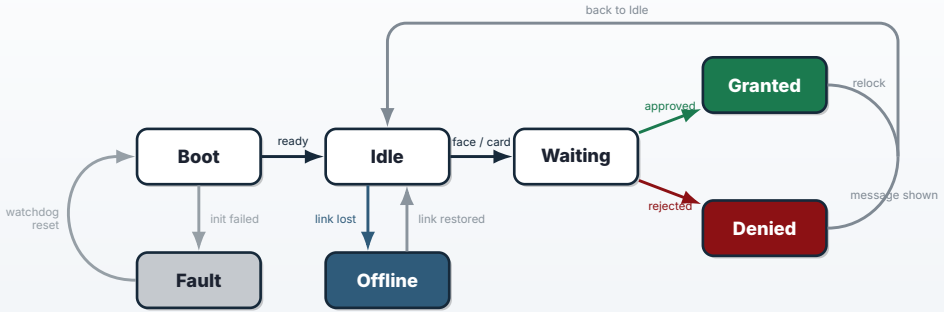
- The node looks the UID up in its **local card store**, synchronized from the server – it does not ask the server on every tap (FR-10a).
- Response in **under 0.5 seconds**, online or offline (PR-2).

When the link is down

- Events are **buffered locally** and carry the time they actually occurred, not the time they were uploaded (FR-20, FR-21).
- They are deleted from local storage **only after the server confirms** each one (FR-22).

Face access is **not available** while offline (FR-24) – only the card path works without a network.

Device state machine



Exactly one state unlocks the latch: **Granted**. Every other state keeps the door shut.

Safety and known limitations

Fail-locked by design (SR-6)

The latch returns to **locked** under every fault: power loss, a software crash, network failure, a peripheral failure, or a corrupt local card store.

Known limitations

- **No presentation-attack detection** (SR-7) – a photograph of an enrolled person may be enough to obtain access.
- RFID card UIDs are **not secret** and can be cloned (SR-11).

Both limitations are stated in the project documentation – a site that needs protection against them must not rely on face recognition as its sole access control.

Thank You

Acknowledgements

Full detail lives in `ARCHITECTURE.md` (design) and `SYSTEM_REQUIREMENTS.md` (requirements) of the LOCK_HYBRID_AI repository.